

Nancy Sotero Silva

PHD STUDENT · UNIVERSITÄT BIELEFELD

282 Dornberger Straße, 33619, Bielefeld, Germany

☎ (+49)15788840116 | ✉ nancy.sotero.320@gmail.com | 🏠 nancysotero.github.io/ | 📷 nancysotero | 🌐 Nancy Sotero | 🎓 Nancy Sotero Silva

“The mind that opens to a new idea never returns to its original size”

Summary

Audiology and Neuroscience researcher passionate about the use of technologies to improve people's brains through their ears. My background includes training in rehabilitation and research experience, with a keen interest in cognition, communication, musical perception and performance. I am currently working on the integration between the oculomotor and auditory efferent systems, but also trained in cortical and behavioral measurements.

Education

Bielefeld University

Bielefeld, Germany

PHD IN BIOLOGY

July 2023 - Currently

- Dissertation: Understanding audio-visual integration from the perspective of eye movement-related eardrums oscillations (EMREOs).
Supervisor: Prof. Dr. Christoph Kayser

Santos Dumont Institute

Macaíba, Brazil

MSC IN NEUROENGINEERING; RESIDENCY ON PERSON WITH DISABILITIES HEALTH CARE

March 2019 - February 2023

- Master's thesis: Influence of stimulus complexity on cortical electrophysiological responses in adults with normal hearing through multichannel acquisition.
Supervisor: Prof. Dr. Edgard Morya

Federal University of Paraíba

João Pessoa, Brazil

BSC IN SPEECH, LANGUAGE AND HEARING SCIENCES

April 2014 - July 2018

- Bachelor's thesis: “Absolute Pitch” in music graduation students: is there a relationship with temporal auditory processing?
Supervisor: Prof. Dr. Hannalice Gottschalck Cavalcanti and Dr. Raphael Bender Chagas Leite

Research Experience

Cognitive Neuroscience Lab, Bielefeld University

Bielefeld, Germany

SCIENTIFIC WORKER/DOCTORAL RESEARCH PROJECT

July 2023 - Currently

- Conducts in-ear recordings (using BioSemi EEG system), eye tracking (EyeLink), hearing tests and psychophysics experiments (PsychToolbox); as long as data curation, processing (FieldTrip), analysis and reporting.

Medical Image Processing Lab, EPFL

Geneva, Switzerland

MASTER'S RESEARCH INTERNSHIP

July - August 2022

- 2 months internship - Summer Research Program.
- Explored processing pipeline optimization (MATLAB) for fNIRS data (Artinis) elicited by cognitive and auditory stimuli. Inspected data from open databases, tested fNIRS processing tools, prepared presentation and written reports.

Santos Dumont Institute

Macaíba, Brazil

MASTER'S AND RESIDENCY RESEARCH PROJECT

2019 - 2023

- Project: Electrophysiological, vascular/hemodynamic and pupillometric measures in cortical plasticity monitoring of adult hearing aids users. Data collection patients from a hearing rehabilitation service aiming the inclusion of follow-up assessments.
- Performed experiment design, data collection and analysis with electroencephalography (BrainVision), ABR, FFR, ERP, pupillometry (Mangold) and fNIRS (NirX), clinical assessment and follow-up of patients (auditory assessment, hearing aids selection and fitting).

Federal University of Paraíba

João Pessoa, Brazil

BACHELOR'S RESEARCH PROJECT

2017 - 2018

- Identified the prevalence of absolute pitch among music undergraduates and its relation to auditory temporal resolution assessment tasks.
- Performed psychophysical experimentation, central auditory processing assessment and music cognition tests; as well as written reports and presentations.

OTHER RESEARCH AND COMMUNITY OUTREACH EXPERIENCES

2015 - 2021

- Communication and hearing assessment and rehabilitation for low-income patients, planning and execution of educational events for the community, organisation of scientific events, reports and presentations.
 - Transdisciplinary assessment of patients complaining of tinnitus
 - Speech therapy for patients with fluency disorders
 - Impact of noise on workers' health

Publication list

1. Wilroth, J., Sotero Silva, N., Tafakkor, A., Mesquita, B., Lau, B., Ip, E., Hannah, J. & Di Liberto, G. (2026). **fNIRS for continuous speech: considerations for hyperscanning**. bioRxiv (preprint), <https://doi.org/10.64898/2026.03.20.7>
2. Sotero Silva, N., Bröhl, F. & Kayser, C. (2025) **Sound lateralization Ability is affected by saccade direction but not EMREOs**. bioRxiv (preprint), <https://doi.org/10.1101/2025.11.05.686724>
3. Sotero Silva, N., Kayser, C. & Bröhl, F. (2025). **Unraveling eye movement-related eardrum oscillations (EMREOs): how saccade direction and tympanometric measurements relate to their amplitude and time course**. Hearing Research, 467. <https://doi.org/10.1016/j.heares.2025.109276>
4. Cavalcanti, H.G., Lucena, M.H.M.S.L., Morya, E., Sotero Silva, N. & Paiva S.F. (2024) **Interface between Music and Audiology** (Chapter). In: Delgado, I.C., Andrade, K.C.L. & Rosa, M.R.D. (ed.). Speech Therapy Intervention in Hearing and Language. 1ed. Rio de Janeiro: Thieme Revinter, p. 56-66. ISBN: 978-65-5572-283-3 [available in portuguese]
5. Sotero Silva, N., Evangelista, C.K.S., Cavalcanti, P.F., Balen, S.A., Cavalcanti, H.G. & Morya, E. (2022). **Review of Audiological Findings in SARS-CoV-2 Infection**. Journal of Neurological Disorders, 10(8). <https://doi.org/10.4172/2329-6895.10.9.514>
6. Pereira, K.V.S., Batista, E.K.L., Souza, P.R.L., Araújo, R.J.S., Sotero Silva, N., Gonçalves, V.S.B., Soares, J.F.R. & Andrade, W.T.L. (2022). **Non-auditory effects of workers exposed to noise and chemicals in a shoe industry**. Brazilian Journal of Development, 8(11), p. 75090–75099. <https://doi.org/10.34117/bjdv8n11-292>
7. Araújo, R.J.S., Sotero Silva, N., Batista, E.K.L., Gonçalves, V.S.B., Soares, J.F.R. & Andrade, W.T.L. (2022). **Impact of noise and toluene on workers' hearing in an footwear industry**. Research, Society and Development, 11(2), e58211225972-e58211225972. <https://doi.org/10.33448/rsd-v11i2.25972>
8. Sotero Silva, N., Batista, E.K.L., Araújo, R.J.S., Moura, K.D., Gonçalves, V.S.B., Soares, J.F.R. & Andrade, W.T.L. (2020) **Auditive complaints in workers exposed to noise and chemicals in footwear industry**. Brazilian Journal of Health Review, 3(3), 6488–6501. <https://doi.org/10.34119/bjhrv3n3-197>
9. Leite Neto, J.T., Silva, L.F.C., Silva, L.J.M., Sotero Silva, N. & Araújo, R.J.S. (2017) **Tinnitus: a general introduction** (Chapter). In: Rosa, M.R.D., Martins, M.L. & Ferreira, T.N.M.(ed.). Multidisciplinary in favour of tinnitus. 1. ed. Pará de Minas: VirtualBooks Editora, p. 16-28. e-ISBN 978-85-434-1182-8 [available in portuguese]

MANUSCRIPTS IN PREPARATION

1. Sotero Silva, N., Vasconcelos, I.C., Alves, E.R.A., Monteiro, T.A., Spinelli, B.G., Miranda, D.C.S., Canto-Pereira, V.R., Balen, S.A. & Morya, E. (2026) **Auditory evoked potentials: does stimulus complexity influence responses?** In preparation
2. Campos, L.G.N., Sotero Silva, N., Alves, E.R.A., Silva, A.R.X., Soares, A.K.A., Dias, R.C.T., Brasil, F.L., Morya, E. & Rosa MRD. (2026) **Transcranial direct current stimulation reduces tinnitus discomfort**. In preparation.
3. Alexandre, V.G.A., Sotero Silva, N. & Vasconcelos, L.D.S. (2026) **Neuropsychology in speech therapy** (Chapter). In: Pinto, P.V.N., Vasconcelos, L.D.S. & Silva, F.T.M. (ed.) History of speech therapy in Brazil. 1. ed. Carapicuíba: Pró-fono. In press. [available in portuguese]

Presentations - Selected

ORAL PRESENTATIONS

2025	47th ECVP - European Conference on Visual Perception , The EMREOs time course differs between ipsi- and contralateral saccades	Mainz, Germany
2024	5th Workshop on Cognitive Neuroscience of Auditory and Cross-modal Perception , Comparing EMREOs induced by different sensory targets	Košice, Slovakia
2023	9th Neuroengineering Symposium , Comparing different stimulation complexities on LLAEPs for temporal and central cortical areas in adults	Macaíba, Brazil
2023	SIG Re(h)abilitat - Universities Telemedicine Network , Speech Therapy and Neuroengineering	Virtual
2022	3rd VCCA - Virtual Conference on Computational Audiology , Electrophysiological and pupillometric measures in monitoring auditory cortex plasticity and listening effort post hearing aids	Virtual
2022	Rehabilitation Colloquium, McGill University , SARS-CoV-2 infection and audiological findings	Virtual

POSTERS

2026	Vision Augmented Hearing, Royal Society , EMREOs during noise and head-tilts [presenter]	Manchester, UK
2025	8th ICAC - International Conference on Auditory Cortex , Relation of EMREOs amplitude and time course with saccade direction and tympanometric measurements [presenter]	Maastricht, The Netherlands
2025	10th ISAAR - International Symposium on Auditory and Audiological Research , EMREOs repercussions on auditory spatial localization [presenter]	Nyborg, Denmark
2024	ESSEM - European Summer School on Eye Movements , EMREOs induced by auditory-guided saccades in different modalities of sound presentation [presenter]	Athens, Greece
2024	FENS Forum , EMREOs are induced by both visual- and auditory-guided saccades [presenter]	Vienna, Austria
2023	38th EIA - International Audiology Meeting , Pupillometric measurements and their relationship to auditory effort & Cognitive evoked potential in normal-hearing adults: active and passive oddball paradigms	Florianópolis, Brazil
2023	9th Neuroengineering Symposium , Impacts of noise exposure in hearing: a longitudinal electroacoustic characterization pilot study in rats	Macaíba, Brazil
2022	8th Neuroengineering Symposium , fNIRS data elicited by motor or auditory stimuli: Processing pipeline and tools for analysis feature extraction [presenter]	Macaíba, Brazil
2017	15th SLAN - Latin American Society of Neuropsychology , Musical perception in cochlear implant users: literature review [presenter] & Anatomy of the musical brain: perfect pitch [presenter]	Natal, Brazil

Awards and Honors

2025	CNSP Hackathon , Awarded 2nd best participant for the work “fNIRS hyperscanning for singing and speaking”	Maastricht, The Netherlands
2025	10th ISAAR, William Demant Foundation , Stipend for attendance and presentation	Nyborg, Denmark
2024	FENS, IBRO-PERC - Early Career Program Award , Stipend for a 3 weeks internship	Budapest, Hungary
2024	EPFL - Summer Research Program , Stipend for a 2 months internship	Budapest, Hungary
2022	8th Neuroengineering Symposium, Santos Dumont Institute , Awarded one of the five best presentations	Macaíba, Brazil
2022	3rd VCCA , Awarded “Video Pitch Award”	Virtual
2018	Federal University of Paraíba , Awarded “Academic Laurea”, graduation with honours in BSc	João Pessoa, Brazil

Skills

Research, Data acquisition and analysis: Electrophysiology (EEG, ERP, ABR, FFR), fNIRS, Eyetracking (Pupillometry, Saccades). Programming: Python (basic), MATLAB (intermediate), LaTeX. Scientific Writing.

Languages, English (C1), Portuguese (native), Spanish (B2), German (A2). Beginner in French, Italian and Brazilian Sign Language.

Teaching and Mentorship

2021-2022	Temporary Lecturer , Otoneurology, Balance Disorders and Occupational Audiology courses, UFRN	Natal, Brazil
2024-2026	Tutor/Lectures , Multisensory Perception and Perception and Action research modules, Bielefeld University	Bielefeld, Germany

Referees

Prof. Dr. Christoph Kayser PhD supervisor (2023 - current). Professor, Chair of Cognitive Neuroscience, Faculty of Biology, Bielefeld University, Bielefeld, Germany.

E-mail: christoph.kayser@gmail.com , christoph.kayser@uni-bielefeld.de

Tel. +49 (521) 106-5700

Dr. Norbert Böddeker PhD second supervisor (2023 - current). Senior Researcher at Cognitive Neuroscience, Faculty of Biology, Bielefeld University, Bielefeld, Germany.

E-mail: nboeddeker@gmail.com , norbert.boeddeker@uni-bielefeld.de

Tel. +49 (521) 106-5712

Prof. Dr. Edgard Morya Master's degree and Residency supervisor (2019 - 2023). Research Coordinator and Professor at Edmond and Lily Safra International Institute of Neuroscience, Santos Dumont Institute, Macaíba, Brazil.

E-mail: edgard.morya@isd.org.br

Tel. +55 (84) 99427-8247